

Combinatorial Solving with Provably Correct Results

Lecture 2

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Topics in Algorithms and Complexity

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October 6, 2026



VERIPB Proof Configuration

Core set $\mathcal{C}$

- Contains input formula at the start
- Input constraints can be modified (e.g., during preprocessing)
- But “equivalence” with input formula maintained

Derived set $\mathcal{D}$

- All constraints derived during search
- Also intermediate constraints used in proof logging [but not used by solver]
- Any satisfying assignment to $\mathcal{C}$ can be extended to $\mathcal{D}$

Objective $f = \sum_i w_i \ell_i + k$

- 0–1 linear function to minimize
- Or $f = 0$ for decision problem
- Keep track of best known bound; initialise to ∞

RUP for Pseudo-Boolean Constraints: Example

Constraint D is **RUP** if adding its negation $\neg D$ and unit propagating yields conflict

Is $D \doteq \overline{geq}_{11} + \overline{lt}_{11} \geq 1$ (i.e., $geq_{11} \Rightarrow \overline{lt}_{11}$) RUP with respect to these constraints?

$$C_1 \doteq 11\overline{geq}_{11} + 8b_3 + 4b_2 + 2b_1 + b_0 \geq 11 \quad \text{slack} = 0$$

$$C_2 \doteq 5\bar{t}_{11} + 8\bar{b}_3 + 4\bar{b}_2 + 2\bar{b}_1 + \bar{b}_0 \geq 5 \quad \text{slack} = -1 \quad \textcolor{red}{\text{⚡}}$$

$$\neg D \doteq \textcolor{brown}{geq}_{11} + \textcolor{brown}{lt}_{11} \geq 2 \quad \text{slack} = 0$$

Unit propagation **to true** and **to false**:

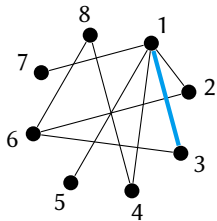
$$geq_{11} \stackrel{-D}{=} 1, lt_{11} \stackrel{-D}{=} 1, b_3 \stackrel{C_1}{=} 1, b_2 \stackrel{C_2}{=} 0, b_1 \stackrel{C_1}{=} 1, b_0 \stackrel{C_1}{=} 1, \perp \stackrel{C_2}{=}$$

Conflict, so D is **RUP** – in VERIPB syntax: `rup 1 ~geq11 1 ~lt11 >= 1 ;`

Topics '26 10/78

More Advanced Example of Redundance-Based Strengthening

Is the following graph 2-colourable (bipartite)?



$$b_1 + b_2 \geq 1$$

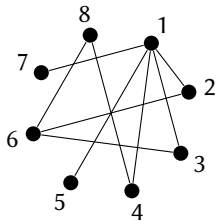
$$\bar{b}_1 + \bar{b}_2 \geq 1$$

$$b_1 + b_3 \geq 1$$

$$\bar{b}_1 + \bar{b}_3 \geq 1$$

More Advanced Example of Redundance-Based Strengthening

Is the following graph 2-colourable (bipartite)?



$$b_1 + b_2 \geq 1$$

$$\bar{b}_1 + \bar{b}_2 \geq 1$$

$$b_1 + b_3 \geq 1$$

$$\bar{b}_1 + \bar{b}_3 \geq 1$$

...

Without loss of generality, node 1 is blue (otherwise we can swap the two colours)

Derive $b_1 \geq 1$ with redundancy; witness
 $\omega : b_i \mapsto \bar{b}_i$ for all i

$$F \doteq F \upharpoonright_\omega \quad \text{and} \quad \neg(b_1 \geq 1) = (b_1 \geq 1) \upharpoonright_\omega$$

pseudo-Boolean proof version 3.0

f 18 ;

```
red 1 b1 >= 1 : b1 -> ~b1 b2 -> ~b2
```

↪ $b_3 \rightarrow \sim b_3 \quad b_4 \rightarrow \sim b_4 \quad b_5 \rightarrow \sim b_5$

↪ $b_6 \rightarrow \sim b_6 \quad b_7 \rightarrow \sim b_7 \quad b_8 \rightarrow \sim b_8 ;$

```
rup >= 1 ;
```

```
output NONE ;
```

```
conclusion UNSAT ;
```

```
end pseudo-Boolean proof ;
```

- Start the proof
- Derive $b_1 \geq 1$ with redundance rule

Status is undetermined

Check provided assignment against original formula

If no assignment given, then

- solution should have been logged
- no unchecked deletion must have occurred before this

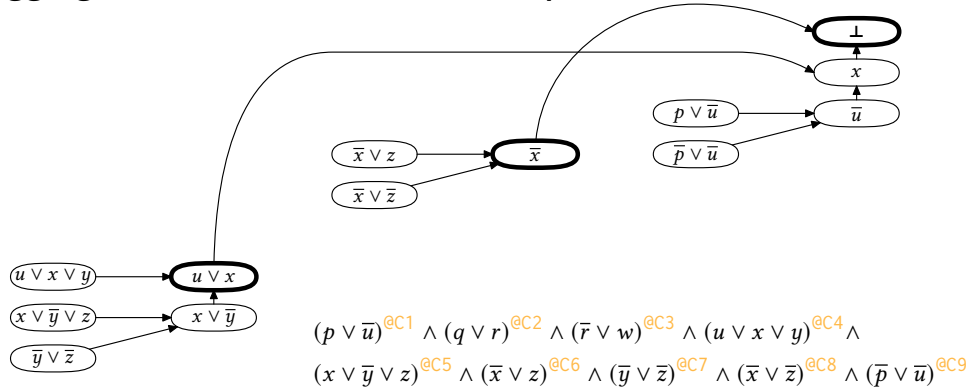
Only valid if no solution has been logged

Check that specified constraint is contradictory (technically: negative slack)

If no constraint specified, check that database contains contradictory constraint



Proof Logging with Labels for CDCL Example



```
@C10 pol @C7 @C5 + 2 d @C4 + 2 d ;
```

$$\rightsquigarrow u + x \geq 1$$

@C11 pol @C8 @C6 + 2 d ;

$$\rightsquigarrow \bar{x} \geq 1$$

@C12 pol @C9 @C1 + 2 d @C10 + 2 d @C11 + 2 d ;

$$\rightsquigarrow 0 \geq 1 \quad \text{⚡}$$

Reification in VERIPB Syntax: Concrete Examples

For $C \doteq x_1 + 2x_2 + 3x_3 + 4x_4 \geq 5$ we have

- $\deg(C) = 5$
- $W = \sum_i a_i = 10$
- $\neg C \doteq \bar{x}_1 + 2\bar{x}_2 + 3\bar{x}_3 + 4\bar{x}_4 \geq 6$

Syntactic sugar	Meaning
$r \implies 1 \ x_1 \ 2 \ x_2 \ 3 \ x_3 \ 4 \ x_4 \geq 5 \ ;$	$5\bar{r} + x_1 + 2x_2 + 3x_3 + 4x_4 \geq 5$
$r \Leftarrow 1 \ x_1 \ 2 \ x_2 \ 3 \ x_3 \ 4 \ x_4 \geq 5 \ ;$	$6r + \bar{x}_1 + 2\bar{x}_2 + 3\bar{x}_3 + 4\bar{x}_4 \geq 6$
$r \Leftrightarrow 1 \ x_1 \ 2 \ x_2 \ 3 \ x_3 \ 4 \ x_4 \geq 5 \ ;$	1st and 2nd constraint, in that order
$r_1 \sim r_2 \ r_3 \implies 1 \ x_1 \ 2 \ x_2 \ 3 \ x_3 \ 4 \ x_4 \geq 5 \ ;$	$5\bar{r}_1 + 5r_2 + 5\bar{r}_3 + x_1 + 2x_2 + 3x_3 + 4x_4 \geq 5$

Still with $C \doteq x_1 + 2x_2 + 3x_3 + 4x_4 \geq 5$, so that $\neg C \doteq \bar{x}_1 + 2\bar{x}_2 + 3\bar{x}_3 + 4\bar{x}_4 \geq 6$
(and recall that $\ell + \bar{\ell} = 1$, so that opposing literals cancel and leave 1)

$$\begin{aligned} C + (r \Leftarrow C) &\doteq 6r + (x_1 + \bar{x}_1) + 2(x_2 + \bar{x}_2) + 3(x_3 + \bar{x}_3) + 4(x_4 + \bar{x}_4) \geq 11 \\ &\doteq 6r \geq 11 - 10 \\ &\doteq \text{deg}(\neg C) \cdot r \geq 1 \end{aligned}$$

The Same Derivation Without Assumptions

Now forget about the assumptions and run exactly the same derivation on the original constraints:

$$\begin{array}{rcl}
 \text{Add} & \frac{r_1 + x_1 + x_2 \geq 1}{2r_1 + 2x_1 + x_2 + x_3 \geq 2} & \frac{r_1 + x_1 + x_3 \geq 1}{x_2 + x_3 \geq 1} \\
 \text{Add} & \frac{2r_1 + 2x_1 + x_2 + x_3 \geq 2}{2r_1 + 2x_1 + 2x_2 + 2x_3 \geq 3} & \\
 \text{Divide by 2} & \frac{2r_1 + 2x_1 + 2x_2 + 2x_3 \geq 3}{r_1 + x_1 + x_2 + x_3 \geq 2} & \\
 \text{Add} & \frac{r_1 + x_1 + x_2 + x_3 \geq 2}{r_1 + \bar{r}_2 \geq 1} & \bar{r}_2 + \bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2
 \end{array}$$

The Same Derivation Without Assumptions

Now forget about the assumptions and run exactly the same derivation on the original constraints:

$$\begin{array}{rcl}
 & r_1 + x_1 + x_2 \geq 1 & r_1 + x_1 + x_3 \geq 1 \\
 \text{Add} & \hline
 & 2r_1 + 2x_1 + x_2 + x_3 \geq 2 & x_2 + x_3 \geq 1 \\
 \text{Add} & \hline
 & 2r_1 + 2x_1 + 2x_2 + 2x_3 \geq 3 & \\
 \text{Divide by 2} & \hline
 & r_1 + x_1 + x_2 + x_3 \geq 2 & \bar{r}_2 + \bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2 \\
 \text{Add} & \hline
 & & r_1 + \bar{r}_2 \geq 1
 \end{array}$$

This yields

$$r_1 + \bar{r}_2 \geq 1 \quad [\text{VERIPB syntax: } 1 \ r1 \ 1 \ \sim r2 \ >= \ 1 \ ;]$$

or equivalently

$$\bar{r}_1 \wedge r_2 \Rightarrow (0 \geq 1) \quad [\text{VERIPB syntax: } \sim r1 \ r2 ==> >= 1 ;]$$

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$$\begin{array}{rcl}
 & r_1 + x_1 + x_2 \geq 1 & r_1 + x_1 + x_3 \geq 1 \\
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 & 2r_1 + 2x_1 + x_2 + x_3 \geq 2 & x_2 + x_3 \geq 1 \\
 \text{Add} & \hline
 & 2r_1 + 2x_1 + 2x_2 + 2x_3 \geq 3 & \\
 \text{Divide by 2} & \hline
 & r_1 + x_1 + x_2 + x_3 \geq 2 & \bar{r}_2 + \bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2 \\
 \text{Add} & \hline
 & & r_1 + \bar{r}_2 \geq 1
 \end{array}$$

This yields

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or equivalently

$$\bar{r}_1 \wedge r_2 \Rightarrow (0 \geq 1) \quad [\text{VERIPB syntax: } \sim r1 \ r2 ==> >= 1 ;]$$

That is, our assumptions imply contradiction

Derivations Under Assumptions: The General Claim

This works in general!

- Suppose that we have
 - 1 constraints $C_1, C_2, \dots, C_m$
 - 2 assumptions $\rho = \{\ell_1 = 1, \ell_2 = 1, \dots, \ell_n = 1\}$
 - 3 cutting planes derivation of D from the restricted constraints $C_1 \upharpoonright_\rho, C_2 \upharpoonright_\rho, \dots, C_m \upharpoonright_\rho$
(i.e., derivation without strengthening)

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(i.e., derivation without strengthening)
- Then the same derivation applied to $C_1, C_2, \dots, C_m$ yields a constraint at least as strong as

$$\ell_1 \wedge \ell_2 \wedge \cdots \wedge \ell_n \Rightarrow D \doteq \deg(D) \cdot \sum_{i=1}^n \bar{\ell}_i + D$$

Derivations Under Assumptions: The General Claim

This works in general!

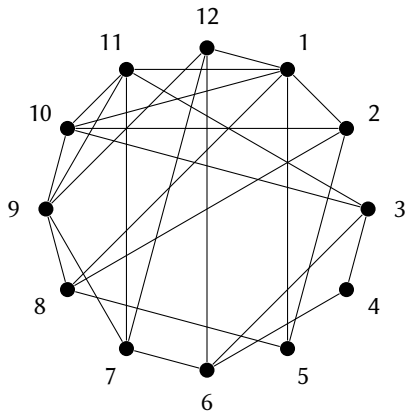
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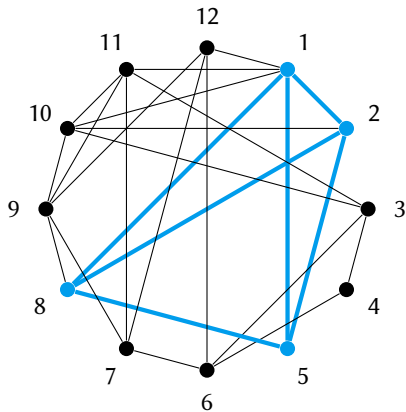
$$\ell_1 \wedge \ell_2 \wedge \cdots \wedge \ell_n \Rightarrow D \doteq \deg(D) \cdot \sum_{i=1}^n \bar{\ell}_i + D$$

Give a convenient way to construct derivations:

- Apply assumptions in ρ to constraints C_j and simplify to get $C_j \upharpoonright_\rho$

Coming next, some worked examples from dedicated graph solvers





There are a lot of dedicated solvers for clique problems

But there are issues:

- “State-of-the-art” solvers have been buggy.
- Often undetected: error rate of around 0.1% [MPP19]

Often used inside other solvers

- An off-by-one result can cause much larger errors

- Pick a vertex v
- Either v is in the clique...
 - Throw away every vertex not adjacent to v
 - If vertices remain, recurse
- ...or v is not in the clique
 - Throw away v (and all incident edges) and pick another vertex

- Growing clique C
- Shrinking set of potential vertices P
 - All the vertices we haven't thrown away yet
 - Every $v \in P$ is adjacent to every $w \in C$

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- Remember the biggest clique C^* found so far
- If $|C| + |P| \leq |C^*|$, no need to keep going

We can use $|C| + \#colours(P)$ as a bound, for any colouring

- 1 Output a pseudo-Boolean encoding of the problem
 - Clique problems have several standard file formats
 - Give the constraints labels, so the proof can refer to them by name
- 2 Make solver generate proof log for search tree
 - Output a small header
 - Output something on every backtrack
 - Output something every time a solution is found
 - Output a small footer
- 3 Figure out how to certify the bound function

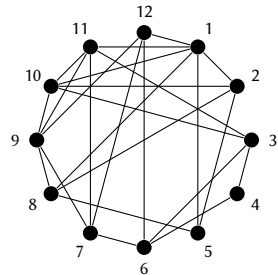
```
* #variable= 12 #constraint= 41
min: 1 ~x1 1 ~x2 1 ~x3 1 ~x4 . . . and so on . . . 1 ~x11 1 ~x12 ;
@noedge1_3 -1 x3 -1 x1 >= -1 ;
@noedge2_3 -1 x3 -1 x2 >= -1 ;
@noedge1_4 -1 x4 -1 x1 >= -1 ;
* . . . and a further 38 similar lines for the remaining non-edges
```

VERIPB proof minimises number of vertices **not** selected, so 4-clique yields objective value $12 - 4 = 8$

First Attempt at a Proof

pseudo-Boolean proof version 3.0

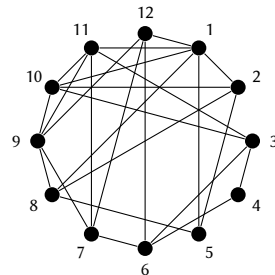
```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```



First Attempt at a Proof

pseudo-Boolean proof version 3.0

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@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
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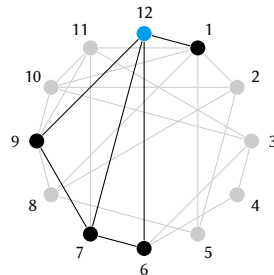
Start with a header

The problem constraints are loaded for us

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
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rup 1 ~x11 >= 1 ;
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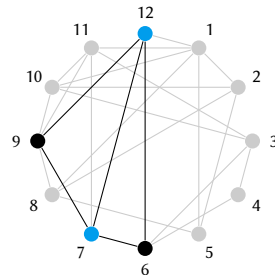


Branch accepting 12
Throw away non-adjacent vertices

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
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rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```

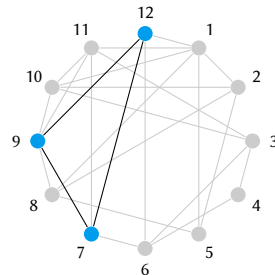


Branch also accepting 7
Throw away non-adjacent vertices

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```

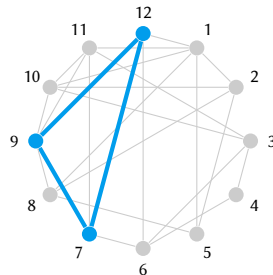


Branch also accepting 9
Throw away non-adjacent vertices

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```



We branched on 12, 7, 9
Found a new incumbent
Now looking for a ≥ 4 vertex clique
“@obj” names the constraint this creates

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
```

```
rup 1 ~x12 1 ~x7 >= 1 ;
```

```
rup 1 ~x12 >= 1 ;
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rup 1 ~x11 1 ~x10 >= 1 ;
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rup 1 ~x11 >= 1 ;
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@obj soli x1 x2 x5 x8 ;
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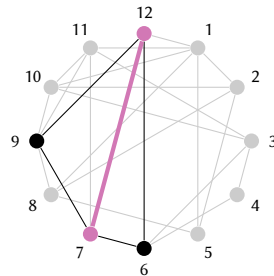
```
rup 1 ~x8 >= 1 ;
```

```
rup >= 1 ;
```

```
output NONE ;
```

```
conclusion BOUNDS 8 8 ;
```

```
end pseudo-Boolean proof ;
```



Backtrack from 12, 7

9 explored already, only 6 feasible

No ≥ 4 vertex clique possible

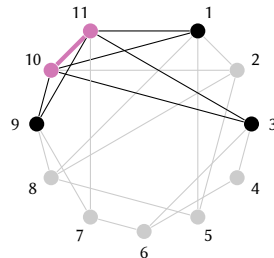
Effectively this deletes the 7–12 edge

Effectively this deletes vertex 12

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```

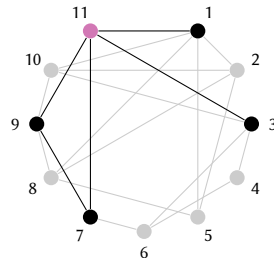


Branch on 11 then 10
Only 1, 3 and 9 feasible (1-colourable)
No ≥ 4 vertex clique possible
Backtrack, deleting the edge

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```

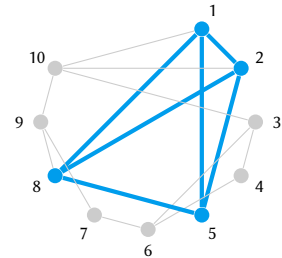


Backtrack from 11
2-colourable, so no ≥ 4 clique
Delete the vertex

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
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rup >= 1 ;
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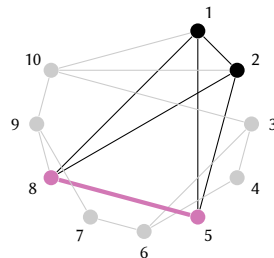


Branch on 8, 5, 1, 2
Find a new incumbent
Now looking for a ≥ 5 vertex clique

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
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rup 1 ~x11 >= 1 ;
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rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
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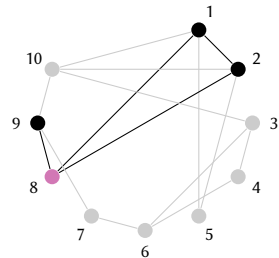


Backtrack from 8, 5
Only 4 vertices; can't have a ≥ 5 clique
Delete the edge

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
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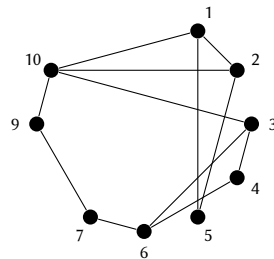


Backtrack from 8
Still not enough vertices
Delete the vertex

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
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rup >= 1 ;
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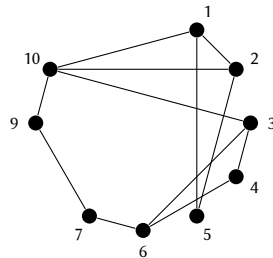
Remaining graph is 3-colourable
Backtrack from root node

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
```

```
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```



- Finish with what we've concluded
- We specify a lower and an upper bound
- Remember we're minimising the number of vertices we *don't* select, so a 4-clique has an objective value of $12 - 4 = 8$
- ("output" is for features we're not using)

Verifying This Proof (Or Not...)

```
$ veripb clique.opb clique-attempt-one.pbp
```

```
Running VeriPB version 3.0.2
```

```
Error: Checking error at clique-attempt-one.pbp:5
```

Caused by:

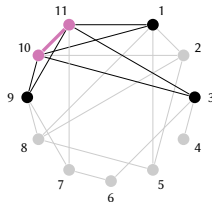
```
The constraint is not implied by reverse unit propagation  
(RUP) from core and derived database. Rerun with the option  
'--trace-failed' to see the propagations. . . .
```

Verifying This Proof (Or Not...)

```
$ veripb clique.opb clique-attempt-one.pbp
Running VeriPB version 3.0.2
Error: Checking error at clique-attempt-one.pbp:5
```

Caused by:

The constraint is not implied by reverse unit propagation (RUP) from core and derived database. Rerun with the option '--trace-failed' to see the propagations. ...



Verifying This Proof (Or Not...)

```
$ veripb --trace clique.opb clique-attempt-one.pbp
Running VeriPB version 3.0.2
  Objective: min 1 ~x1 1 ~x2 . . . 1 ~x11 1 ~x12 0
  ConstraintId 1: 1 ~x1 1 ~x3 >= 1
  ConstraintId 2: 1 ~x2 1 ~x3 >= 1
  . . .
  ConstraintId 41: 1 ~x11 1 ~x12 >= 1
line   1: pseudo-Boolean proof version 3.0
line   2: @obj soli x7 x9 x12 ;
  ConstraintId 42: 1 x1 1 x2 . . . 1 x11 1 x12 >= 4
line   3: rup 1 ~x12 1 ~x7 >= 1 ;
  ConstraintId 43: 1 ~x7 1 ~x12 >= 1
line   4: rup 1 ~x12 >= 1 ;
  ConstraintId 44: 1 ~x12 >= 1
line   5: rup 1 ~x11 1 ~x10 >= 1 ;
Error: Checking error at clique-attempt-one.pbp:5
```

Verifying This Proof (Or Not...)

```
$ veripb --trace-failed clique.opb clique-attempt-one.pbp
Running VeriPB version 3.0.2
Propagation check failed! The propagation had the following trail:
  propagations in format: <assignment> (<reason constraint>)
    ~x12 (1 ~x12 >= 1)
    x10 (1 x10 1 x11 >= 2)
    x11 (1 x10 1 x11 >= 2)
    ~x4 (1 ~x4 1 ~x10 >= 1)
    ~x5 (1 ~x5 1 ~x10 >= 1)
    ~x6 (1 ~x6 1 ~x10 >= 1)
    ~x7 (1 ~x7 1 ~x10 >= 1)
    ~x8 (1 ~x8 1 ~x10 >= 1)
    ~x2 (1 ~x2 1 ~x11 >= 1)
Error: Checking error at clique-attempt-one.pbp:5
```

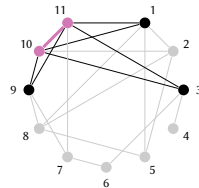
The Problem of Counting

The proof checker isn't smart enough to work out that

- vertices 1, 3 and 9 can all be given the same colour...
- and each colour class contributes at most one vertex to a clique...
- so “find me a clique with more than three vertices” can't be satisfied once we've accepted both 10 and 11

Unit propagation never sees this:

- The bound needs **counting**
- But the encoding only ever talks about **pairs** of vertices



Dealing With Colourings

So the colour bound doesn't follow by reverse unit propagation...

But we can lazily recover at-most-one constraints for each colour class!

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But we can lazily recover at-most-one constraints for each colour class!

$$\begin{aligned}
 &(\bar{x}_1 + \bar{x}_6 \geq 1) \\
 + &(\bar{x}_1 + \bar{x}_9 \geq 1) &= 2\bar{x}_1 + \bar{x}_6 + \bar{x}_9 \geq 2 \\
 + &(\bar{x}_6 + \bar{x}_9 \geq 1) &= 2\bar{x}_1 + 2\bar{x}_6 + 2\bar{x}_9 \geq 3 \\
 & &/ 2 &= \bar{x}_1 + \bar{x}_6 + \bar{x}_9 \geq 2 \\
 & & &\text{i.e. } x_1 + x_6 + x_9 \leq 1
 \end{aligned}$$

Dealing With Colourings

So the colour bound doesn't follow by reverse unit propagation...

But we can lazily recover at-most-one constraints for each colour class!

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 &(\bar{x}_1 + \bar{x}_6 \geq 1) \\
 + &(\bar{x}_1 + \bar{x}_9 \geq 1) &&= 2\bar{x}_1 + \bar{x}_6 + \bar{x}_9 \geq 2 \\
 + &(\bar{x}_6 + \bar{x}_9 \geq 1) &&= 2\bar{x}_1 + 2\bar{x}_6 + 2\bar{x}_9 \geq 3 \\
 &&&/ 2 &&= \bar{x}_1 + \bar{x}_6 + \bar{x}_9 \geq 2 \\
 &&&&&\text{i.e. } x_1 + x_6 + x_9 \leq 1
 \end{aligned}$$

This generalises to colour classes of any size v

- Each non-edge is used exactly once, $v(v-1)$ additions
- $v-3$ multiplications and $v-2$ divisions

Solvers don't need to "understand" cutting planes to write out this derivation

Details on At-Most-One Constraint Recovery (3 Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
- I.e., $\bar{x}_1 + \dots + \bar{x}_n \geq n - 1$ from constraints $\bar{x}_i + \bar{x}_j \geq 1$ in normalised form
- Need this to prove that **at most one vertex per colour class** can be a member of the clique

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$$\text{Add} \quad \frac{\bar{x}_1 + \bar{x}_2 \geq 1 \quad \bar{x}_1 + \bar{x}_3 \geq 1}{2\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2}$$

Details on At-Most-One Constraint Recovery (3 Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
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- Need this to prove that **at most one vertex per colour class** can be a member of the clique

$$\begin{array}{rcl}
 & \bar{x}_1 + \bar{x}_2 \geq 1 & \bar{x}_1 + \bar{x}_3 \geq 1 \\
 \text{Add} & \hline
 & 2\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2 & \bar{x}_2 + \bar{x}_3 \geq 1 \\
 \text{Add} & \hline
 & 2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 3
 \end{array}$$

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- Need this to prove that **at most one vertex per colour class** can be a member of the clique

$$\begin{array}{rcl}
 & \bar{x}_1 + \bar{x}_2 \geq 1 & \bar{x}_1 + \bar{x}_3 \geq 1 \\
 \text{Add} & \hline
 & 2\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2 & \bar{x}_2 + \bar{x}_3 \geq 1 \\
 \text{Add} & \hline
 & 2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 3 \\
 \text{Divide by 2} & \hline
 & \bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq \frac{3}{2}
 \end{array}$$

Details on At-Most-One Constraint Recovery (3 Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
- I.e., $\bar{x}_1 + \dots + \bar{x}_n \geq n - 1$ from constraints $\bar{x}_i + \bar{x}_j \geq 1$ in normalised form
- Need this to prove that **at most one vertex per colour class** can be a member of the clique

$$\begin{array}{rcl}
 & \bar{x}_1 + \bar{x}_2 \geq 1 & \bar{x}_1 + \bar{x}_3 \geq 1 \\
 \text{Add} & \hline
 & 2\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2 & \bar{x}_2 + \bar{x}_3 \geq 1 \\
 \text{Add} & \hline
 & 2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 3 \\
 \text{Divide by 2} & \hline
 & \bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq \left\lceil \frac{3}{2} \right\rceil
 \end{array}$$

Details on At-Most-One Constraint Recovery (3 Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
- I.e., $\bar{x}_1 + \dots + \bar{x}_n \geq n - 1$ from constraints $\bar{x}_i + \bar{x}_j \geq 1$ in normalised form
- Need this to prove that **at most one vertex per colour class** can be a member of the clique

$$\begin{array}{rcl}
 & \bar{x}_1 + \bar{x}_2 \geq 1 & \bar{x}_1 + \bar{x}_3 \geq 1 \\
 \text{Add} & \hline
 & 2\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2 & \bar{x}_2 + \bar{x}_3 \geq 1 \\
 \text{Add} & \hline
 & 2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 3 \\
 \text{Divide by 2} & \hline
 & \bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2
 \end{array}$$

Details on At-Most-One Constraint Recovery (3 Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
- I.e., $\bar{x}_1 + \dots + \bar{x}_n \geq n - 1$ from constraints $\bar{x}_i + \bar{x}_j \geq 1$ in normalised form
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$$\begin{array}{rcl}
 & \bar{x}_1 + \bar{x}_2 \geq 1 & \bar{x}_1 + \bar{x}_3 \geq 1 \\
 \text{Add} & \hline
 & 2\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2 & \bar{x}_2 + \bar{x}_3 \geq 1 \\
 \text{Add} & \hline
 & 2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 3 \\
 \text{Divide by 2} & \hline
 & \bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2
 \end{array}$$

Assuming constraint $\bar{x}_i + \bar{x}_j \geq 1$ has label @C*ij*, get VERIPB code

```
pol @C12 @C13 + @C23 + 2 d ;
```

Details on At-Most-One Constraint Recovery (4 and More Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
- I.e., $\bar{x}_1 + \dots + \bar{x}_n \geq n - 1$ from constraints $\bar{x}_i + \bar{x}_j \geq 1$ in normalised form
- Need this to prove that **at most one vertex per colour class** can be a member of the clique

$$\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2$$

Details on At-Most-One Constraint Recovery (4 and More Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
- I.e., $\bar{x}_1 + \dots + \bar{x}_n \geq n - 1$ from constraints $\bar{x}_i + \bar{x}_j \geq 1$ in normalised form
- Need this to prove that **at most one vertex per colour class** can be a member of the clique

$$\text{Multiply by 2} \quad \frac{\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2}{2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 4}$$

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$$\begin{array}{rcl}
 \text{Multiply by 2} & \frac{\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2}{2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 4} & \bar{x}_1 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 4}{3\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 + \bar{x}_4 \geq 5} & \bar{x}_2 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{3\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 + \bar{x}_4 \geq 5}{3\bar{x}_1 + 3\bar{x}_2 + 2\bar{x}_3 + 2\bar{x}_4 \geq 6} & \bar{x}_3 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{3\bar{x}_1 + 3\bar{x}_2 + 2\bar{x}_3 + 2\bar{x}_4 \geq 6}{3\bar{x}_1 + 3\bar{x}_2 + 3\bar{x}_3 + 3\bar{x}_4 \geq 7} &
 \end{array}$$

Details on At-Most-One Constraint Recovery (4 and More Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
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 \text{Multiply by 2} & \frac{\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2}{2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 4} & \bar{x}_1 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 4}{3\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 + \bar{x}_4 \geq 5} & \bar{x}_2 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{3\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 + \bar{x}_4 \geq 5}{3\bar{x}_1 + 3\bar{x}_2 + 2\bar{x}_3 + 2\bar{x}_4 \geq 6} & \bar{x}_3 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{3\bar{x}_1 + 3\bar{x}_2 + 2\bar{x}_3 + 2\bar{x}_4 \geq 6}{3\bar{x}_1 + 3\bar{x}_2 + 3\bar{x}_3 + 3\bar{x}_4 \geq 7} & \\
 \text{Divide by 3} & \frac{3\bar{x}_1 + 3\bar{x}_2 + 3\bar{x}_3 + 3\bar{x}_4 \geq 7}{\bar{x}_1 + \bar{x}_2 + \bar{x}_3 + \bar{x}_4 \geq \left\lceil \frac{7}{3} \right\rceil} &
 \end{array}$$

Details on At-Most-One Constraint Recovery (4 and More Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
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 \text{Add} & \frac{3\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 + \bar{x}_4 \geq 5}{3\bar{x}_1 + 3\bar{x}_2 + 2\bar{x}_3 + 2\bar{x}_4 \geq 6} & \bar{x}_3 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{3\bar{x}_1 + 3\bar{x}_2 + 2\bar{x}_3 + 2\bar{x}_4 \geq 6}{3\bar{x}_1 + 3\bar{x}_2 + 3\bar{x}_3 + 3\bar{x}_4 \geq 7} & \\
 \text{Divide by 3} & \frac{3\bar{x}_1 + 3\bar{x}_2 + 3\bar{x}_3 + 3\bar{x}_4 \geq 7}{\bar{x}_1 + \bar{x}_2 + \bar{x}_3 + \bar{x}_4 \geq 3} &
 \end{array}$$

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- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
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 \text{Add} & \frac{3\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 + \bar{x}_4 \geq 5}{3\bar{x}_1 + 3\bar{x}_2 + 2\bar{x}_3 + 2\bar{x}_4 \geq 6} & \bar{x}_3 + \bar{x}_4 \geq 1 \\
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 \text{Divide by 3} & \frac{3\bar{x}_1 + 3\bar{x}_2 + 3\bar{x}_3 + 3\bar{x}_4 \geq 7}{\bar{x}_1 + \bar{x}_2 + \bar{x}_3 + \bar{x}_4 \geq 3} &
 \end{array}$$

Assuming constraint $\bar{x}_i + \bar{x}_j \geq 1$ has label @C ij , VERIPB code for full derivation is

```
pol @C12 @C13 + @C23 + 2 d 2 * @C14 + @C24 + @C34 + 3 d ;
```

Details on At-Most-One Constraint Recovery (4 and More Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
- I.e., $\bar{x}_1 + \dots + \bar{x}_n \geq n - 1$ from constraints $\bar{x}_i + \bar{x}_j \geq 1$ in normalised form
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$$\begin{array}{rcl}
 \text{Multiply by 2} & \frac{\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2}{2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 4} & \bar{x}_1 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{\quad}{3\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 + \bar{x}_4 \geq 5} & \bar{x}_2 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{\quad}{3\bar{x}_1 + 3\bar{x}_2 + 2\bar{x}_3 + 2\bar{x}_4 \geq 6} & \bar{x}_3 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{\quad}{3\bar{x}_1 + 3\bar{x}_2 + 3\bar{x}_3 + 3\bar{x}_4 \geq 7} & \\
 \text{Divide by 3} & \frac{\quad}{\bar{x}_1 + \bar{x}_2 + \bar{x}_3 + \bar{x}_4 \geq 3} &
 \end{array}$$

Assuming constraint $\bar{x}_i + \bar{x}_j \geq 1$ has label @C ij , VERIPB code for full derivation is

```
pol @C12 @C13 + @C23 + 2 d 2 * @C14 + @C24 + @C34 + 3 d ;
```

(and this pattern generalizes to any number of variables)

What The Revised Proof with Explicit Counting Looks Like

pseudo-Boolean proof version 3.0

@obj soli x7 x9 x12 ;

rup 1 ~x12 1 ~x7 >= 1 ;

pol @noedge1_6 @noedge1_9 + @noedge6_9 + 2 d @obj + ;

rup 1 ~x12 >= 1 ;

pol @noedge1_3 @noedge1_9 + @noedge3_9 + 2 d @obj + ;

rup 1 ~x11 1 ~x10 >= 1 ;

pol @noedge1_3 @noedge1_7 + @noedge3_7 + 2 d @obj + ;

rup 1 ~x11 >= 1 ;

@obj soli x1 x2 x5 x8 ;

rup 1 ~x8 1 ~x5 >= 1 ;

pol @obj @noedge1_9 + ;

rup 1 ~x8 >= 1 ;

pol @noedge1_3 @noedge1_7 + @noedge3_7 + 2 d @noedge2_4 @noedge2_9 + @noedge4_9 + 2 d +
 @noedge5_6 @noedge5_10 + @noedge6_10 + 2 d + @obj + ;

rup >= 1 ;

output NONE ;

conclusion BOUNDS 8 8 ;

end pseudo-Boolean proof ;

What The Revised Proof with Explicit Counting Looks Like

pseudo-Boolean proof version 3.0

@obj soli x7 x9 x12 ;

rup 1 ~x12 1 ~x7 >= 1 ;

pol @noedge1_6 @noedge1_9 + @noedge6_9 + 2 d @obj + ;

rup 1 ~x12 >= 1 ;

pol @noedge1_3 @noedge1_9 + @noedge3_9 + 2 d @obj + ;

rup 1 ~x11 1 ~x10 >= 1 ;

pol @noedge1_3 @noedge1_7 + @noedge3_7 + 2 d @obj + ;

rup 1 ~x11 >= 1 ;

@obj soli x1 x2 x5 x8 ;

rup 1 ~x8 1 ~x5 >= 1 ;

pol @obj @noedge1_9 + ;

rup 1 ~x8 >= 1 ;

pol @noedge1_3 @noedge1_7 + @noedge3_7 + 2 d @noedge2_4 @noedge2_9 + @noedge4_9 + 2 d +

@noedge5_6 @noedge5_10 + @noedge6_10 + 2 d + @obj + ;

rup >= 1 ;

output NONE ;

conclusion BOUNDS 8 8 ;

end pseudo-Boolean proof ;

Failing RUP steps have become **pol** derivations: add three non-adjacency constraints together, divide by two, then add the result to the solution-improving constraint

The labels come straight out of the OPB file

What The Revised Proof with Explicit Counting Looks Like

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
pol @noedge1_6 @noedge1_9 + @noedge6_9 + 2 d @obj + ;
rup 1 ~x12 >= 1 ;
pol @noedge1_3 @noedge1_9 + @noedge3_9 + 2 d @obj + ;
rup 1 ~x11 1 ~x10 >= 1 ;
pol @noedge1_3 @noedge1_7 + @noedge3_7 + 2 d @obj + ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
pol @obj @noedge1_9 + ;
rup 1 ~x8 >= 1 ;
pol @noedge1_3 @noedge1_7 + @noedge3_7 + 2 d @noedge2_4 @noedge2_9 + @noedge4_9 + 2 d +
    @noedge5_6 @noedge5_10 + @noedge6_10 + 2 d + @obj + ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```

Three colour classes here, so three at-most-one constraints, all added to the solution-improving constraint

Verifying This Proof (For Real, This Time)

```
$ veripb clique.opb clique-new-attempt.pbp
Running VeriPB version 3.0.2
s VERIFIED BOUNDS 8 <= obj <= 8
```

No errors this time — every step is justified

Note that the checker **knows nothing** about

- cliques,
- colourings, or
- bound functions

Verifying This Proof (For Real, This Time)

```
$ veripb --trace clique.opb clique-new-attempt.pbp
Running VeriPB version 3.0.2
Objective: min 1 ~x1 1 ~x2 . . . 1 ~x11 1 ~x12 0
ConstraintId 1: 1 ~x1 1 ~x3 >= 1
. . .
ConstraintId 41: 1 ~x11 1 ~x12 >= 1
line 2: @obj soli x7 x9 x12 ;
ConstraintId 42: 1 x1 1 x2 . . . 1 x11 1 x12 >= 4
line 3: rup 1 ~x12 1 ~x7 >= 1 ;
ConstraintId 43: 1 ~x7 1 ~x12 >= 1
line 4: pol @noedge1_6 @noedge1_9 + @noedge6_9 + 2 d @obj + ;
ConstraintId 44: 1 x2 1 x3 1 x4 1 x5 1 x7 1 x8 1 x10 1 x11 1 x12 >= 3
. . .
line 16: rup >= 1 ;
ConstraintId 55: >= 1
s VERIFIED BOUNDS 8 <= obj <= 8
```

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Why not just put a “colouring bound” rule into the proof format?

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- Even more for [maximum weight clique](#), where weighted colour class rules get extremely clever (vertices can split their weight between several colours)

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Cutting planes proof system to the rescue!

- Turns out to be strong enough for all of these cases
- Proof length scales linearly with solver reasoning time (asymptotically)

Different Clique Algorithms

Different search orders?

- ✓ Irrelevant for proof logging

Using local search to initialise?

- ✓ Just log the incumbent

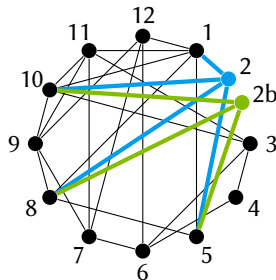
Different bound functions?

- Is cutting planes strong enough to justify every useful bound function ever invented?
- So far, seems like it...

Weighted cliques?

- ✓ Multiply a colour class by its largest weight
- ✓ Also works for vertices “split between colour classes”

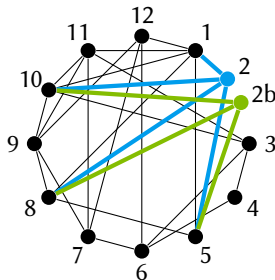
Lazy Global Domination for Maximum Clique [MP16]



Can ignore vertex 2b

- Every neighbour of 2b is also a neighbour of 2
- This is an example of **dominance** reasoning

Lazy Global Domination for Maximum Clique [MP16]



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Dominance-based strengthening rule can justify this

- Even when detected dynamically during search

Reducing the Trust Base

We are still trusting rather a lot

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- VERIPB can **elaborate** a proof into a simpler **kernel format**
- Uses only rules that the **formally verified** checker **CAKEPB** knows about
- The proof checker CAKEPB (which is written in CAKEML and proved correct in HOL4)
 - reads the **original graph file**
 - does a **formally verified translation to 0–1 ILP**
 - performs **formally verified checking** of the kernel proof against this translation

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 - reads the **original graph file**
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- So the answer is formally verified in terms of the original input

End-to-End Verification for Maximum Clique

```
$ glasgow_clique_solver brock200_4.clq
recursions = 56613
clique = 17 (12 19 28 29 38 54 65 71 79 93 117 127 139 161 165 186 192)
runtime = 0.02153s

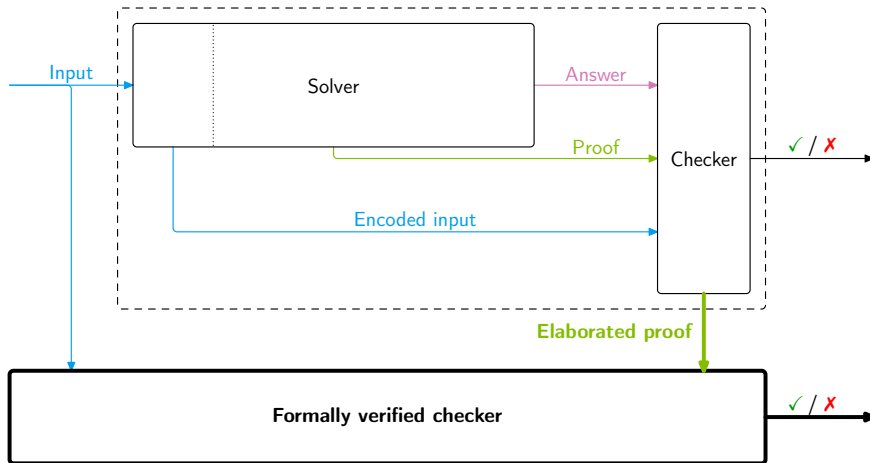
$ glasgow_clique_solver brock200_4.clq --prove brock200_4
runtime = 0.266475s

$ cake_pb_clique brock200_4.clq > brock200_4.verifiedopb

$ veripb --elaborate brock200_4.corepb brock200_4.verifiedopb brock200_4.pbp
s VERIFIED BOUNDS 183 <= obj <= 183

$ cake_pb_clique brock200_4.clq brock200_4.corepb
s VERIFIED MAX CLIQUE SIZE |CLIQUE| = 17
```

Workflow for Solving with End-to-End Certified Results



What's Left to Trust?

Still have to trust:

- The HOL4 theorem prover
- That the formal model of the CAKEML environment matches the hardware it runs on
- The HOL4 definition of what it means to be a maximum clique
- Input parsing and output formatting

What's Left to Trust?

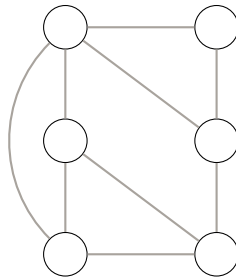
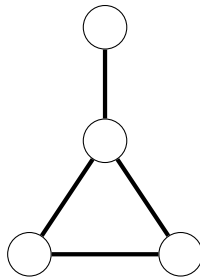
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No need to trust, or even to know about:

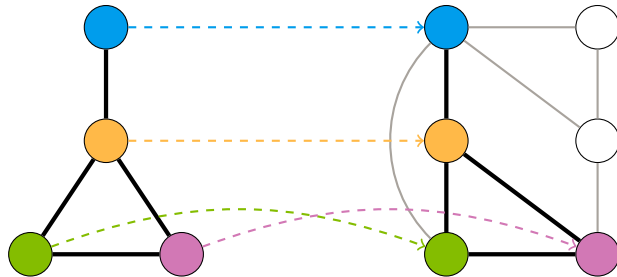
- How the solver works
- That the VERIPB proof system is sound
- Or even what “pseudo-Boolean” means

Subgraph Isomorphism



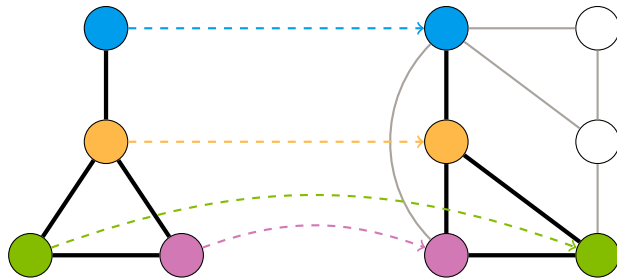
- Find the **pattern** inside the **target**
- Applications in compilers, biochemistry, model checking, pattern recognition, ...
- Often want to find **all** matches

Subgraph Isomorphism



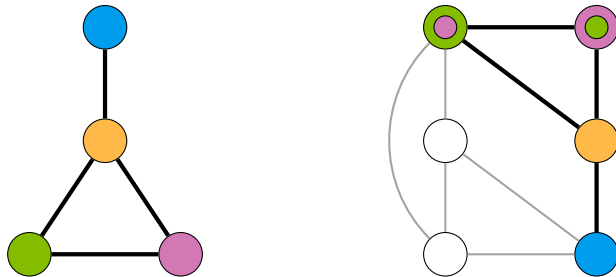
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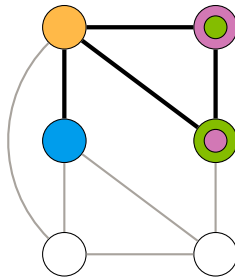
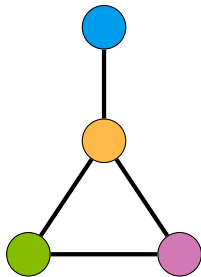
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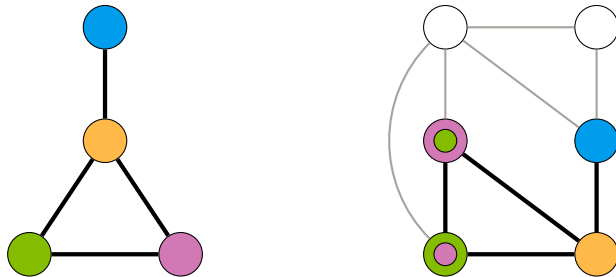
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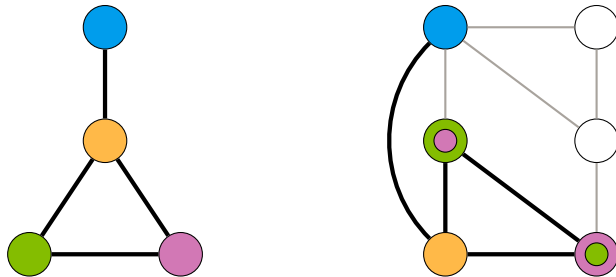
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Subgraph Isomorphism in Pseudo-Boolean Form

Each pattern vertex gets a target vertex:

$$\sum_{t \in V(T)} x_{p,t} = 1 \qquad p \in V(P)$$

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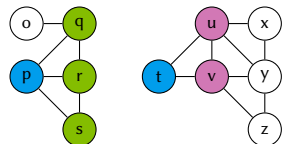
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Adjacency constraints: if p maps to t , then neighbours of p map to neighbours of t :

$$\bar{x}_{p,t} + \sum_{u \in N(t)} x_{q,u} \geq 1 \quad p \in V(P), q \in N(p), t \in V(T)$$

Degree Reasoning in Cutting Planes



Pattern vertex p of degree $\deg(p)$ can never be mapped to target vertex t of degree $< \deg(p)$

Observe $N(p) = \{q, r, s\}$ and $N(t) = \{u, v\}$

We wish to derive $\bar{x}_{p,t} \geq 1$

$$3 \cdot \bar{x}_{p,t} \geq 1$$

Degree Reasoning in VERIPB

```

pol @adj_p_t_q @adj_p_t_r + @adj_p_t_s +      % sum adjacency constraints
    @inj_u + @inj_v +                          % sum injectivity constraints
    xo_u + xo_v +                              % cancel stray xo_* >= 0
    xp_u + xp_v +                              % cancel stray xp_* >= 0
    3 d ;                                       % divide, and we're done

```


Degree Reasoning in VERIPB

```

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    @inj_u + @inj_v +                          % sum injectivity constraints
    xo_u + xo_v +                             % cancel stray xo_* >= 0
    xp_u + xp_v +                             % cancel stray xp_* >= 0
    3 d ;                                       % divide, and we're done

```

The labels come from the OPB file, so solver doesn't need to keep track of constraint IDs

Or we can let VERIPB do the last bit of tidying up for us:

```

pol @adj_p_t_q @adj_p_t_r + @adj_p_t_s +      % sum adjacency constraints
    @inj_u + @inj_v + ;                       % sum injectivity constraints
rup 1 ~xp_t >= 1 ;                           % implied by the above

```

Proof Logging for Other Forms of Reasoning

Possible to certify (almost) all of the other things state-of-the-art subgraph solvers do:

- Conditional injectivity reasoning and filtering
- Distance filtering
- Neighbourhood degree sequences
- Path filtering
- Supplemental graphs

Limitations

Why trust the encoding?

- ✓ For clique and subgraph isomorphism this is now taken care of [GMM⁺24]
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Works up to moderately-sized hard instances

- Even an $O(n^3)$ encoding is painful
- Particularly bad when the pseudo-Boolean encoding talks about “non-edges” but large sparse graphs are “easy”

Reformulations?

Integer Variables in Constraint Programming (1/2)

How to deal with integer variables in constraint programming?

Given $A \in \{-3, \dots, 9\}$, the direct encoding is:

$$a_{=-3} + a_{=-2} + a_{=-1} + a_{=0} + a_{=1} + a_{=2} + a_{=3} \\ + a_{=4} + a_{=5} + a_{=6} + a_{=7} + a_{=8} + a_{=9} = 1$$

Integer Variables in Constraint Programming (2/2)

We can mix binary and order encodings! Define big-M linear inequalities encoding

$$a_{\geq 4} \Leftrightarrow -16a_{\text{neg}} + 1a_{\text{b0}} + 2a_{\text{b1}} + 4a_{\text{b2}} + 8a_{\text{b3}} \geq 4$$

$$a_{\geq 5} \Leftrightarrow -16a_{\text{neg}} + 1a_{\text{b0}} + 2a_{\text{b1}} + 4a_{\text{b2}} + 8a_{\text{b3}} \geq 5$$

$$a_{=4} \Leftrightarrow a_{>4} \wedge \bar{a}_{>5}$$

When creating $a_{>i}$, also introduce pseudo-Boolean constraints encoding

$$a_{\geq i} \Rightarrow a_{\geq j} \quad \text{and} \quad a_{\geq h} \Rightarrow a_{\geq i}$$

for the closest values $j < i < h$ that already exist

Constraints can be specified **extensionally** as list of feasible tuples, called a **table**
Variable assignments must match some row in table

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on a

No need to prove anything about guesses/decisions – only justify backtracking

$$V \in \{1 \quad 4 \quad 5\}$$

$$Z \in \{1, 3\} \quad z_{=1} + z_{=3} \geq 1 \quad [Z \text{ takes some value}]$$

$$-v_{=1} \geq 1 \quad [\text{Sum all constraints so far}]$$

Justifying All-Different Failures

$$Z \in \{1, 3, \dots\} \quad z_{=1} + z_{=3} \geq 1 \quad [Z \text{ takes some value}]$$

$$\rightarrow -w_{=3} + -x_{=3} + -y_{=3} + -z_{=3} \geq -1 \quad [\text{At most one variable} = 3]$$

$$v_{=1} \geq 0 \quad [\text{Variable } v_{=1} \text{ non-negative}]$$

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$$v_{=1} \geq 0 \quad [\text{Variable } v_{=1} \text{ non-negative}]$$

$$0 \leq 1 \quad [\text{Sum above two constraints}]$$

Need modelling language with formally specified semantics...

The *Glasgow* Constraint Solver

Support for 37 (families of) constraints (and counting), with all inferences certified:

Abs, AllDifferent, AllEqual, Among, AtMostOne, BinPacking, Circuit, Comparison, Count, Cumulative, Diffn, Disjunctive, DivMod, Element, Equals, GCC, In, Increasing, Inverse, Knapsack, Lex, Linear, Logical, MDD, MinDistance, MinMax, Multiply, NValue, Parity, PlusMinus, Power, Regular, SeqPrecedeChain, SmartTable, Sort, Table, ValuePrecede

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Also **certifies reformulations** not written in but inferred from input:

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Also **certifies reformulations** not written in but inferred from input:

- Auto-tabulation: Identify small subproblem, solve it, and replace by table
- Capacity strengthening by integrality, conflict cliques across resources, lifted cover cuts: Extract information implied by scheduling problem constraints
- Difference logic: Spot that a pile of two-term inequalities is a precedence network, and hand to temporal propagator

Future Research Directions

Performance of pseudo-Boolean proof logging and checking

- Faster proof logging and checking!
- Trim proofs while checking as in SAT solving [HHW13] (*upcoming work [ABB⁺26a]*)
- Compress proof files using binary format

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Proof logging for other combinatorial problems and techniques

- Model counting
- Mixed integer linear programming (*suggested extension of VERIPB in [DEGH23]*)
- Satisfiability modulo theories (SMT) solving (*Lots of work in, e.g., [HS22, BBC⁺23, RSB⁺26, RSS⁺26]*)

And more...

- Use proof logs for algorithm analysis or explainability purposes
 - Lots of other challenging problems and interesting ideas
 - Talk to us if you want to join the pseudo-Boolean proof logging revolution! ☺
- We're happy to **collaborate**, and **we're hiring**

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